JAN–P–491

28 AUGUST 1947

JOINT ARMY-NAVY SPECIFICATION

PARACHUTE-CORD, ARTIFICIAL-FIBER (For Ammunition Parachutes)

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specifications:

U. S. Army	Navy Department
50–11–116A 2 Aug. 1944	.....
50–11–118 2 Dec. 1943	
50–11–123 1 Jan. 1944	
50–11–128 3 Jan. 1944	

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

U. S. ARMY SPECIFICATIONS

50–0–1—General Specification for Ammunition except Small Arms Ammunition.[1]

100–2—Standard Specification for Marking Shipments by Contractors.[1]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

A–2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[2]

B. TYPES AND CLASSES

B–1. This specification covers the following types and classes of artificial fiber cord as specified in the contract or order. (See par. H–2.)

Type A—*Rayon*

Class a— 40 pounds min. breaking load
b— 95
c—125
d—250

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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e—300
f—325
g—400
h—450
i—850
j—900

Type B—*Nylon*

Class a—105 pounds min. breaking load
b—400
c—550

C. MATERIAL AND WORKMANSHIP

C–1. *Yarn.*—Cord shall be manufactured of yarn of sufficiently high quality and tenacity to insure conformance of the finished product with the requirements of this specification. There shall be no knots, lumps, or broken or free ends projecting from the finished cord.

C–2. *Construction.*—

C–2a. *Type A.*—

C–2a(1). *General.*—Type A cord shall be manufactured from rayon.

C–2a(1)a. *Classes a and b.*—Classes a and b cord shall be woven on a braiding machine having not less than 12 and 16 carriers, respectively.

C–2a(1)b. *Classes c, d, f, h, and j.*—Classes c, d, f, h, and j shall consist of an outer sleeve with a central core. The outer sleeve shall be made on a braiding machine having not less than 16 carriers with the exception of class j for which a minimum of 32 carriers is required. The weave and the minimum number of picks per inch for these classes measured under 1 pound tension shall be as shown in table I.

TABLE I.—*Construction of classes c, d, f, h, and j cord.*

No. of carriers on braiding machine	Weave	Picks per inch minimum
16	Standard or 2 under and 2 over	8
16	Basket	4 visible
24	Standard or 2 under and 2 over	10
24	Basket	5 visible
32	Standard or 2 under and 2 over	8 for class j
32	Standard or 2 under and 2 over	12 for classes c, d, f, g, h

Classes c, d, f, h, and j core yarn shall be cabled and even and regular in twist without kinks or other manufacturing imperfections. Construction shall be of a number of ends of yarn being first plied into threads by twisting in one direction and then, with multiples of similarly plied threads, twisted together in the opposite direction to form strands. The core shall be of sufficient

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size to maintain the finished cord in circular form. The core of classes c, d, f, and h shall consist of not more than 4 strands of core yarn and the core of class j shall consist of not more than 8 strands of core yarn.

C-2a(1)c. *Class i.*—Class i cord shall consist of an outer sleeve without core, but shall otherwise conform to the same constructional requirements as class j cord.

C-2a(1)d. *Class g.*—Class g cord shall consist of an outer sleeve without core and shall be braided on a machine with a minimum of 16 carriers using the standard or 2 under and 2 over weave and shall show a minimum of 8 picks per inch under 1 pound tension.

C-2a(1)e. *Class e.*—Class e cord shall be woven on a loom and shall be of plain one up and one down tubular construction or herringbone twill of solid construction using a warp yarn of 275, 300, or 900 denier with a total weight of 88,500 to 93,000 denier. The fill yarn shall be 275 or 300 denier, 28 to 32 picks per inch. The following construction may be used in the manufacture of class e cord, subject to meeting all other requirements specified herein:

900 denier regular yarn in warp—51 face ends, 2 ends in heddle for total of 102 ends or 300 denier regular yarn in warp—51 face ends—6 ends in heddle for total of 306 ends.

Entering draft—8 shafts:
10 ends on shafts 1, 3, 5, 7
10 ends on shafts 2, 4, 6, 8
10 ends on shafts 5, 7, 1, 3
10 ends on shafts 6, 8, 2, 4
11 ends on shafts 1, 3, 5, 7

Weaving draft—8 cards:
Card Nos. 1 and 5, punch out Nos. 1, 5, 6, 7
Card Nos. 2 and 6, punch out Nos. 1, 2, 6, 8
Card Nos. 3 and 7, punch out Nos. 1, 2, 3, 5
Card Nos. 4 and 8, punch out Nos. 2, 4, 5, 6

C-2a(2). *Treatment for moisture repellency.*—Class a cord shall not be treated for moisture repellency. Class b cord shall be treated for moisture repellency. For all other classes, the outer sleeve and core of the braided cord, when manufactured of yarns exceeding 300 denier size, shall be treated for moisture repellency to reduce the rate of moisture absorption. The type of moisture repellent used should be an aluminum acetate wax. The treatment employed shall not reduce the pliability of the cord or immediately or ultimately affect the minimum physical requirements as specified in table II. Treatment for moisture repellency is optional if the cord is manufactured of 300 denier yarn or less.

C-2b. *Type B.*—

C-2b(1). *General.*—Type B cord shall be manufactured from nylon yarn which shall not be subjected to a stretching operation. Sleeve yarn shall be shrunk for a minimum of 30 minutes at 160° ± 5° F. while core yarn shall be shrunk for a minimum of 60 minutes at 200 ± 5° F. The yarn shall then be dried at room temperature before braiding. No oil shall be added. The outer

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sleeves shall be made on a 32 or 36 carrier braiding machine, single end, and shall have not less than 26 nor more than 28 picks per inch; class a cord may be made on a 16 carrier machine, using two ends per carrier, if desired. The sleeve yarn of class a cord shall be either 70 or 210 denier and the sleeve yarn of classes b and c shall be 210/3. Sleeve yarns shall have not less than 7.5 nor more than 9 turns per inch spin twist and not less than 5.0 nor more than 6.0 turns per inch ply twist.

C–2b(2). *Classes a and b.*—Classes a and b cord shall consist of an outer sleeve without core.

C–2b(3). *Class c.*—Class c cord shall consist of an outer sleeve with a central core. The core of class c cord shall consist of seven, eight, or nine threads, each 3 ply, of a size sufficient to maintain the finished cord in a circular form. The core yarns for class c cord shall be made 210/5/3. The first ply of 5 ends shall have not less than 10 nor more than 12 turns per inch, and the final ply shall have not less than 6 nor more than 7 turns per inch.

D. GENERAL REQUIREMENTS

D–1. *Color identification.*—Each manufacturer shall insert not more than 4 colored threads which will identify his product. Where sleeve and core construction is used, the colored threads shall be part of the core, using regular core threads. Otherwise, the colored threads shall be part of the sleeve or solid construction using regular threads. The threads shall be fast dyed and each colored identification thread shall consist of a single yarn and/or plied end. When more than 1 colored thread is required in the identification symbol, the colored identification threads shall be placed in separate plies. The number and colors of the identifying threads will be assigned to individual manufacturers upon application to the procuring agency. Dyed ends shall be considered fast dyed if there is no appreciable alteration of color when the dyed yarns are subjected to dry cleaning and laundering tests as specified in paragraphs F–4f(1) and F–4f(2).

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E. DETAIL REQUIREMENTS

E–1. *Physical properties.*—The completed cord shall have the physical properties shown in table II.

TABLE II.—*Physical properties.*

Designation		Elongation Percent (min.)	Area of cross section Sq. mm. (max.)	Yards per pound (min.)	Breaking load Pounds (min.)
Type	Class				
A	a	[1]	1.1	550	40
	b	15	4.00	200	95
	c	20	4.5	140	125
	d	20	9	70	250
	e	18[2]	[3]	40	300
	f	20	10	60	325
	g	20	[4]	40	400
	h	20	16	40	450
	i	18	[5]	18	850
	j	20	35	18	900
B	a	30	..	310	105
	b	30	..	90	400
	c	30	..	75	550

[1] At breaking strength, 8.5; at 90-percent breaking strength, 7.5.
[2] Maximum, 25.
[3] Width ½ inch plus 0, minus 1/16 inch. Width x thickness, 0.028 sq. in. max.
[4] Width, 0.40 inch max. Width x thickness, 0.028 sq. in. max.
[5] Width, ½ inch ± 1/16 inch. Width x thickness, 0.068 sq. in. max.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Size of lot.*—Maximum, 1,000 pounds.

F–2. *Sampling.*—Ten of the spools or rolls comprising the lot, shall be selected by the inspector. Approximately 20 yards of cord shall be cut from each of the selected spools, and after conditioning in accordance with paragraph F–4a, shall be cut to provide one specimen for each of the applicable tests under paragraph F–4.

F–3. *Inspection.*—

F–3a. *Army.*—Inspection shall be made in accordance with U. S. Army Specification 50–0–1 and shall be at the point of delivery unless otherwise specified in the contract or order.

F–3b. *Navy.*—Unless otherwise specified in the contract or order, inspection shall be made at the point of delivery.

F–4. *Tests.*—Tests to determine compliance with the requirements of this specification shall be conducted in accordance with the following paragraphs as applicable. For Navy purchases, the tests shall be conducted at a Government laboratory unless otherwise specified in the contract or order.

F–4a. *Atmospheric conditions.*—The test samples of cord shall be placed in an air circulating oven and dried at a temperature of 105° C. to 110° C. (221° F. to 230° F.). Repeat the drying at 105° C. until a constant weight is obtained. Immediately after attaining the dry weight, the samples shall be conditioned for 4 hours in a standard atmosphere of 70° F. temperature, with a

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tolerance of plus 10° F., and 65 percent relative humidity, with a tolerance of plus or minus 2 percent, prior to performing physical tests as specified in paragraphs F–4b, F–4c, F–4d, and F–4e. All physical tests shall be performed while in above standard atmosphere.

F–4b. *Breaking load.*—Cut a test specimen 6 feet in length from the sample. It is preferable that the breaking load of these specimens be determined by means of a pendulum-type machine of such a capacity that the arm makes an angle between 4° and 45° with the vertical when the rupture occurs. Each grip of the machine shall consist of a steel or iron cylinder not less than 2 inches in diameter and 2 inches in length, and having a polished surface. The cylinders or grips shall be not less than 10 inches between centers at the beginning of the test and shall separate at the rate of 12 inches per minute under no load. In case a specimen breaks less than ½ inch from the point at which it touches the drum grip or at any point on the grip, disregard the result and test another specimen from the sample.

F–4c. *Yards per pound.*—Cut from the sample a specimen 10 yards in length, measured when subjected to only sufficient tension to keep the specimen taut. The tension shall not exceed 2 pounds. Calculate the number of yards per pound.

F–4d. *Elongation.*—Cut a test specimen from the sample. With the test specimen under 0.5 pound tension in the same machine used for determining the breaking load, mark off a 5-inch length between the drums. With calipers or stretch-followers, follow these marks as the tension is increased steadily to rupture. The elongation shall be calculated as the percent increase in length, at the moment of rupture, measured between the marks originally 5 inches apart. Another specimen from the sample shall be tested if the original test is improperly performed.

F–4e. *Area of cross section.*—Measure the undistorted maximum and minimum thicknesses of each piece of the sample while under tension of not more than one pound. A satisfactory method for this measurement utilizes a measuring microscope with a scale graduated in 0.1 mm divisions employing a magnification of 12.7. Consider the area of cross section of the cord to be equivalent to that of a circle of diameter equal to the average of the maximum and minimum thickness. Calculate the area of cross section of the specimen on this basis.

F–4e(1). *Width x thickness.*—With the specimen under a tension of one pound and stretched over a smooth circular surface of at least one inch radius, measure the width and thickness of the specimen at two places using suitable measuring devices such as a scale or micrometer. Calculate the average width x thickness of the specimen on this basis.

F–4f. *Color fastness.*—

F–4f(1). *Dry cleaning.*—Cut a specimen approximately 4 inches in length, with marker thread included, from the sample. The specimen for test shall be placed in a pint jar containing 150 ml. of dry-cleaning solvent and agitated in a Launder-Ometer, or similar machine, for 20 minutes at 90° ± 5° F. The solution in

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the jar shall then be poured out and any discoloration resulting from the bleeding of the specimen shall be noted. The specimen shall be removed from the jar and allowed to dry at room temperature. The test shall be rated in accordance with paragraph F–4f(3).

F–4f(2). *Laundering.*—Cut a specimen approximately 4 inches in length, with marker thread included, from the sample. The specimen for test shall be placed in a pint jar containing 100 ml. of a 0.5-percent soap solution. The jar shall then be tightly closed and clamped in the Launder-Ometer. The jar shall be rotated for 30 minutes with the water bath temperature held at 180° F. Remove the specimen from the jar, rinse first in warm water and then in cold water, wrap in a cloth, and dry. The test shall be rated in accordance with paragraph F–4f(3).

F–4f(3). *Evaluation of color fastness.*—The color of the specimens shall be compared visually with the color of the corresponding unwashed specimen. In making the comparison, the specimens shall be held in the same plane and viewed at the same angle under a standard daylight lamp or equivalent illuminant. A specimen shall be considered to pass the test for color fastness when there is no appreciable change in color, and also no appreciable staining of the undyed portion of the specimen. "Appreciable change" shall be defined as staining which is immediately noticeable without necessitating a comparison of the tested specimen with a portion of the undyed cord. Should closer inspection be required to make apparent the change in color or the amount of staining, the change shall not be considered to be "appreciable."

F–5. *Rejection and resubmission.*—If any specimen of the sample fails to meet all the requirements of sections D and E, the lot shall be rejected. Rejected lots of cord may be resubmitted after having been reworked provided that the method of reworking is acceptable to the contracting officer. Twenty spools or rolls shall be selected from the resubmitted lot as samples, and all tests specified in paragraph F–4 shall be performed.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging.*—

G–1a. *Type A.*—

G–1a(1). Classes a and b cord shall be wound on spools holding 2 to 4 pounds of material. Each turn and layer shall be free of entanglement.

G–1a(2). Classes c, d, and f cord shall, unless otherwise specified, be furnished on spools containing 215 yards for classes d and f, and 500 yards for class c. A spool shall contain not more than three pieces. Each piece shall be at least 20 yards in length. Ends shall be laid loose on spools and not fastened together.

G–1a(3). Classes e, i, and g cord shall, unless otherwise specified, be furnished on spools containing 200 yards for classes e and i, and 100 yards for class g. Each spool shall consist of not more than five pieces with no one piece less than 8 yards in length. Ends on a spool shall be laid loose and shall not be fastened together in any way. Scotch tape splicing shall not be used. Each

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piece and the bottom end of the spool shall have a warning flag of heavy red yarn 8 yards before the end of the piece as the cord is unrolled from the spool.

G–1a(4). Class h cord shall, unless otherwise specified, be furnished on spools containing 215 yards in one piece. However, a number of spools, not to exceed 10 percent of the spools in any lot, may contain two pieces with a minimum piece length of 16 yards. Class j cord shall, unless otherwise specified, be furnished on spools containing 115 yards in one piece. However, a number of spools, not to exceed 10 percent of the spools in any lot, may contain two pieces with a minimum piece length of 14 yards. Ends shall be laid loose on spools and not fastened together. Spools shall be marked to indicate the number of pieces if containing more than one piece.

G–1b. *Type B.*—

G–1b(1). Class a cord shall be furnished on spools containing 500 to 600 yards. A spool shall contain not more than three pieces and no piece shall be less than 20 yards in length. Ends shall be laid loose on spools and not fastened together.

G–1b(2). Classes b and c cord shall be furnished on spools each containing 220 yards in one piece, except that not over 10 percent of the spools in any lot may contain two pieces with a minimum piece length of 16 yards. Ends shall be laid loose on the spools and not fastened together. Each separate piece of type A cord and the bottom end of the spool shall have a warning flag of heavy red yarn 8 yards before the end of the piece as the cord is unrolled in use from the spool.

G–2. *Packing.*—Unless otherwise specified, the spools of cord shall be properly separated by type and class, and packed in commercial containers so constructed as to insure acceptance by common or other carriers for safe transportation at the lowest rate, to the point of delivery.

G–3. *Marking.*—In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook.

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H. NOTES

H–1. *Use.*—This specification covers artificial-fiber cord for use in ammunition parachutes.

H–2. Requests, requisitions, schedules, and contracts or orders should specify the following:

(a) Title, number, and date of the specification.

(b) Type and class required. (See par. B–1.)

H–3. This specification replaces Ordnance Department, U. S. Army Tentative Specification AXS–1463.

H–4. Copies of Joint Army-Navy specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy and United States Army specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–5. Copies of Joint Army-Navy specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–6. Copies of this Joint Army-Navy specification (required for Army purchases) may be obtained from the Office, Chief of Ordnance, War Department, Washington 25, D. C.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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